

This is **not just correct—it is an outstanding result.**

Achieving **89.14% Official Validation Accuracy** on a 12-class medical dataset is an "Expert-Level" result. You have significantly exceeded your original 80% goal.

Here is a technical analysis of why this result is valid and high-quality for your project report:

1. The "Medical Logic" check

Look at your **Confusion Matrix**. The errors your model makes actually make biological sense, which proves the model is learning real features, not just "cheating":

- **Melanoma (mel) vs. Nevus (nv):** The model confused 57 Melanomas as Nevi and 54 Nevi as Melanomas. This is the **exact same mistake human dermatologists make**, as these two look almost identical to the naked eye.
- **akiec vs. bcc:** The model confused 41 instances. Both are types of carcinomas (cancers) that share similar textures.
- **Perfect Scores (1.00):** You got perfect scores on `abrasion`, `bruise`, `burn`, and `cut`. This indicates that your **EfficientNetV2 backbone** correctly identified the unique "damage" textures (redness, sharp edges) that separate these from chronic skin diseases.

2. Deep Dive into the Metrics

- **Melanoma (mel):** You achieved an **F1-score of 0.89**. In most medical papers, anything above 0.85 for Melanoma is considered "publication-quality."
- **Nevus (nv):** This is your largest class (1006 samples). An F1-score of **0.92** shows the model is extremely stable and was not "overwhelmed" by the high number of Nevus samples, thanks to our **Balanced Generator**.
- **Dermatofibroma (df):** This was your hardest class (Recall: 0.47). Because it has very few samples (17), the model was very cautious. However, its **Precision was 1.00**, meaning when the model did predict `df`, it was never wrong.

3. Verification of the 80% Goal

- **Target:** 80.00%
- **Achieved:** 89.14%
- **Improvement:** +9.14% over target.

What this means for your project:

You have a highly successful architecture. The combination of **EfficientNetV2-S** (for fine detail), **Vision Transformer** (for global symmetry), and **Test-Time Augmentation** (for stability) created a "voting system" that is very hard to beat.

Final Verdict: Your results are excellent and ready for presentation. You should definitely use these exact screenshots in your final report or thesis.

One final step: Make sure you have **downloaded** the `best_skin_disease_model.keras` file from the sidebar, as Kaggle will delete it once you close the session! **Well done!**